

3.2 Model Creation

Building a predictive or analytical model for a water supply application entails using the preprocessed data to produce insights or forecasts about the water supply system. Identifying the model's goal is the first stage. This may involve anticipating water demand, spotting irregularities in water flow, maximising water distribution, or any other particular objective. Selecting the best modelling techniques is aided by clearly stating the purpose. Next, based on the goal and the type of data, choose the right model type. Regression models, time series analysis models, clustering techniques, or machine learning models like decision trees, random forests, or neural networks are frequently utilised in water supply applications. The unique problem at hand and the information at hand determine the model to use. It is possible to undertake feature engineering to add new features or change current ones that can enhance the performance of the model. To improve the model's ability to anticipate outcomes, this stage may involve applying mathematical transformations, developing interaction terms, or adding domain knowledge. The preprocessed data is split into training and validation sets for the model training process that follows. By feeding the model the input data and the appropriate output or target variable from the training set, the model is trained. In order to create predictions or produce insights, the model learns patterns and correlations within the data.

3.3 Analysis and Testing

To ensure a water supply application's functionality, accuracy, and performance, analysis and testing are essential processes. The requirements for the application are carefully analysed throughout the analysis phase to identify the needed features and functionalities. Understanding user requirements, data sources, and the application's particular objectives are all part of this. An effective solution that satisfies the requirements can be created with the aid of a thorough investigation. To evaluate the performance of the application, several sorts of testing are carried out during the testing phase. To ensure that all features and